P04: Makers Makin' It, Act II -- The Seequel

Design Doc by RustyRoosters, pd. 5

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PROJECT NAME: Fitness Genie
DESCRIPTION: Our website is a meal calorie calculator. Users can sign up/login, track a meal's calories through a db of known food calories from a dataset, save them to their account, and view them later on a personal profile page. Users can look through charts and graphs to view how their meals compare to a large sample of data.

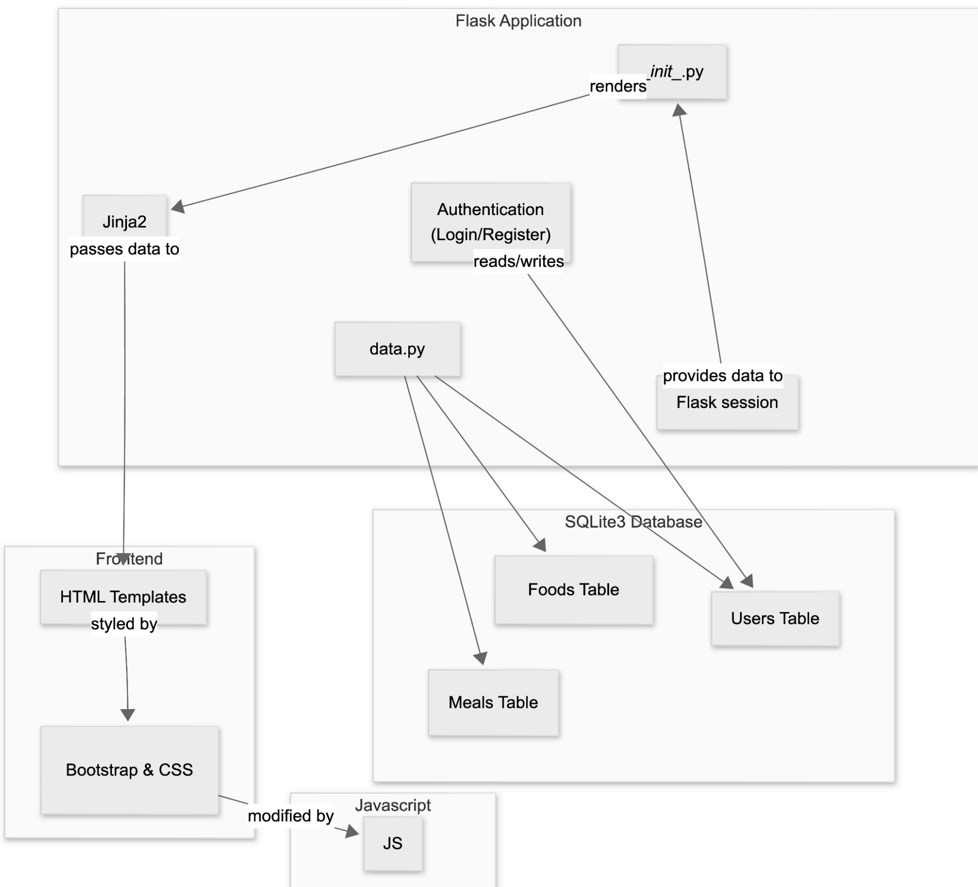
TARGET SHIP DATE: 4/20/2026

Program Components:

1) Flask App (Python)
    a) **__init__.py** (main app + routes)
        i) Creates Flask app, config, and session handling
        ii) Renders templates and connects frontend to backend logic
        iii) Routes:
            (1) **/register** adds a new user to the **users** table (**data.py**). Checks if the username is unique, hashes the password, stores it in session, then redirects to **/dashboard**.
            (2) **/login** checks users (**data.py**), verifies password hash, stores session, redirects to **/dashboard**.
            (3) **/logout** clears session, redirects to **/login**.
            (4) **/home** displays data on the nutrition of different foods as well as general calorie intake of a sample of 2000 people
            (5) **/calculate** allows users to select food that they eat in a day and it will calculate the calorie intake and compare it with the rest of the general data from the nutrition intake dataset by redirecting the to **/visualize**
            (6) **/visualize** Displays interactive graphs using D3.js based on:
                - User's calorie intake compared with general nutrition dataset
                - Macronutrient breakdown (carbs, protein, fat)
                  Users can filter by:
                - Nutrient type, etc.
            (7) **/profile/** uses **data.py** to load a creator's saved meals, calories, and nutrition stats
    b) data.py
        i) Connects to SQLite3 database and creates/maintains tables + parse csv dataset files and add them to the database
2) Database (SQLite3) (stored in data.db)
    a) **users** table stores all usernames and password hashes for authentication.
    b) **foods** nutritional data
    c) **meals** user meal logs
    d) **meal-items** stores which food items in which meal
3) Frontend Framework

    a) **Bootstrap**: used for navbar, forms (login/registration), buttons, layout/grid, and cards for food items to keep styling consistent and responsive.
4) JavaScript
    a) **visualize.js**: renders D3.js graphs and handles data filtering
    b) **dashboard.js**: displays overview data and basic visualizations
5) RESTful APIs
    a) None
6) External CSS (if needed)

Component Map:



Database Organization

USERS			
TEXT	name	PK	Unique
TEXT	password		For authentication
DATE	created_at		

Foods			
INTEGER	id	PK	Auto-increment
TEXT	name		
INTEGER	calories		
FLOAT	protein		(grams)
FLOAT	carbs		(grams)
FLOAT	fat		(grams)

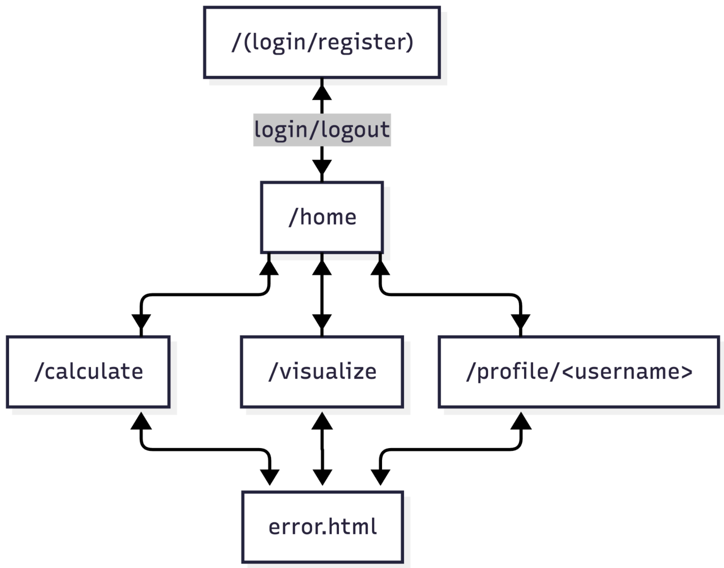
Meals			
INTEGER	id	PK	Auto-increment
TEXT	username	FK	USERS(username)

Meal-items			
INTEGER	id	PK	Auto-increment
INTEGER	meal_id	FK	MEALS(id)
INTEGER	food_id	FK	FOODS(id)
INTEGER	quantity		

Site Map:

7) Templates (HTML)
    a) **/login** (username + password. Error message if login fails).
    b) **/register** (username + password. Error message if username is taken).
    c) **(dashboard.html) /dashboard** (Displays: Overview of nutrition data, basic D3 visualizations, Info about healthy calorie ranges)
    d) (**calculate.html**) **/calculate** Allows users to:
        - Select foods from database
        - Input quantity
        - Calculate total daily calorie intake
        - Compare result with general dataset
    e) (**visualize.html**) **/visualize** Interactive D3 graphs:
        - Line chart (calories over time or vs. average)
        - Bar chart (macronutrient breakdown)
        - Comparison chart (user vs dataset)
    f) (**profile.html**) **/profile/<username>** (shows username, nutritional stats).

g) **error.html** (invalid id's or permission errors).



Visualization Library (D3 JS)

1. Line graphs (calorie intake over time)
2. Scatterplot (maps points for calorie intake of a sample of people)
3. Bar charts (compares macronutrients of meal to sample data)
4. Difference charts (user vs dataset averages)

APIs

(None needed)

Frontend Framework

Bootstrap

Datasets

1. https://www.kaggle.com/datasets/abdussamad123/user-daily-nutritional-intake
   - Contains daily nutritional intake across a large sample of people which we will use to compare to the user's stats
2. https://www.kaggle.com/datasets/utsavdey1410/food-nutrition-dataset/data
   - Contains nutritional values for specific food items to calculate the nutrients in the user's meals

Tasks & Assignments

TASK BREAKDOWN

Task	**Devo(s)**
Login & Register	James Sun
Editor UI & Javascript	Matthew Ciu
DB Organization	Yuhang Pan
D3JS Visualizations	Thomas Mackey